西南财经大学本科期末考试试题册（A）

（2024－2025学年第1学期）

以下各项由命题教师填写：

课程名称：数据结构（英）**命题教师：**陈中普 王海林

适用对象（年级专业）**：**2023级信息管理与信息系统、2023级物流管理、2023级金融数学

使用试题的任课教师姓名：陈中普 王海林

试题说明：

1、考试类型：闭卷[√] 开卷[ ]

2、本套试题共 三 道大题，共 6 页，完卷时间 120 分钟。

3、考试用品中除纸、笔、尺子外，可另带的用具有：

计算器[ ] 字典[ ] 等

（请在下划线上填上具体数字或内容，所选[ ]内打钩）

考试时间（由制卷方填写）**：**

以下各项由学生填写：

任课教师： **年级专业：**

学生姓名： **学 号：**

考生注意事项：1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。

2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。

3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。

4.所有答案均需填写在答题纸上，答在试题册上无效。

5.考生不得携带任何通讯工具进入考场。

6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。

7.严格遵守考场纪律。

- The height of an empty tree is -1.
- The node in a singly linked list has attribute *data* and *next*, in which *data* is the element it is holding, and *next* is the link to the next node. If it is doubly linked list, the node holds *prev* link to the previous nodes.
- The node in a binary tree has attribute *key, left*, and *right*, in which *key* is the data it is holding, and *left* and *right* are links to its left and right child, respectively.

Part One: Multi-choices (2 marks each; 20 marks in total)

(*Note: Each question has only one correct choice.*)

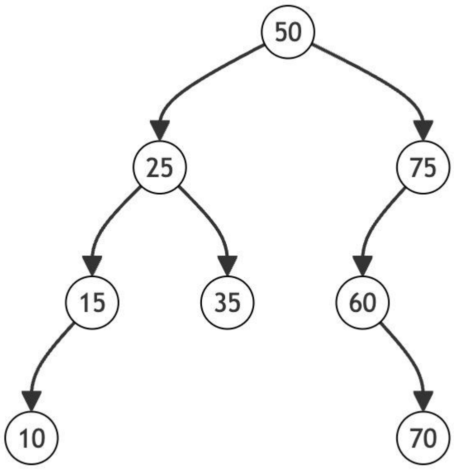
1. Suppose an initially empty stack S has executed a total of 15 push operations, 4 peek operations, and 10 pop operations, in which 4 pop operations have raised Empty errors. What is the size of S?
   A. 5
   B. 6
   C. 8
   D. 9

2. Given an initially empty stack S and empty queue Q. Elements *a, b, c, d, e, f, g* were pushed into S sequentially. If the popped element was be enqueued into Q immediately, and the seven elements dequeued were in the order of *b, d, c, f, e, a, g,* then what is minimal capacity of S?
   A. 3
   B. 4
   C. 5
   D. 6

3. Which of the following statements is NOT correct in terms of big O notation?
   A. The function 16N + 7 can be written as $O(N^2)$.
   B. To search an item in a linked list, the time complexity is O(1) in the best case.
   C. To search an item in an array, the time complexity is O(1) in the best case.
   D. An algorithm in O(N) is always faster than one in $O(N^2)$.

4. Suppose the level of the root is 1. Given a complete binary tree, there are only 8 nodes at level 6. What is the number of nodes in this tree?
   A. 39
   B. 52
   C. 111
   D. 119

5. Given the BST below, after deleting 25 and 75, what is the pre-order walk of it?



A. 50 → 35 → 15 → 10 → 60 → 90
B. 10 → 15 → 35 → 50 → 60 → 90
C. 50 → 10 → 15 → 35 → 90 → 60
D. 10 → 50 → 35 → 15 → 60 → 90

6. Given a max-heap in an array [25, 13, 10, 12, 9]. How many comparisons are required to re-heapify it after inserting 18?
   A. 1
   B. 2
   C. 3
   D. 4

7. What is the time complexity of the following code snippet?

```
x = 4
while x < 5 * N:
    x = 3 * x
```

   A. O(N)
   B. O(N^2)
   C. O(1)
   D. O(*log*N)

8. Given a graph of n vertices and m edges, what is the space overhead when it is represented with adjacent lists?
   A. O(mn)
   B. O(m + n)
   C. O(n^2)
   D. O(m^2)

9. Which of the following is NOT a valid Huffman code for any set of symbols?
   A. {0, 10, 110, 111}
   B. {0, 10, 100, 110}
   C. {00, 01, 10, 11}
   D. {01, 00, 10, 11}

10. Given a hash table of size 11 using linear probing for collision resolution. The hash function is h(key) = key % 11. Insert the following keys in order: 22, 44, 50, 12, 66 What is the final position of 66 in the table?
    A. 1
    B. 2
    C. 3
    D. 4

Part Two: True or False (2 mark each; 20 marks in total)

1. To address the incompatible speed between printers and computers, it is generally to set up a buffer area based on a stack. (True/False)
2. Given a singly linked list with head and tail pointer, the time complexity of removing the tail is O(N). (True/False)
3. If $f(n) = 8n^2 + 5 = O(n^2)$, and $g(n) = O(n^2)$, then $f(n) = g(n)$. (True/False)
4. Given a BST, the first element in its in-order traversal must be the smallest element. (True/False)
5. The search key performance of a BST can be as bad as a singly linked list. (True/False)
6. As for the separating chaining for hash table of size N, a search takes $O(log N + \alpha)$, where α is the load factor. (True/False)
7. A max-heap using an array satisfies the binary search property, so its search time complexity is $O(log N)$. (True/False)
8. DFS can be used to find the shorted path between two vertices. (True/False)
9. It is possible for a black-red tree to contain black nodes only, without any red node. (True/False)
10. To implement a queue using a singly linked list with head and tail pointer, the dequeue can be efficiently implemented by removing head.

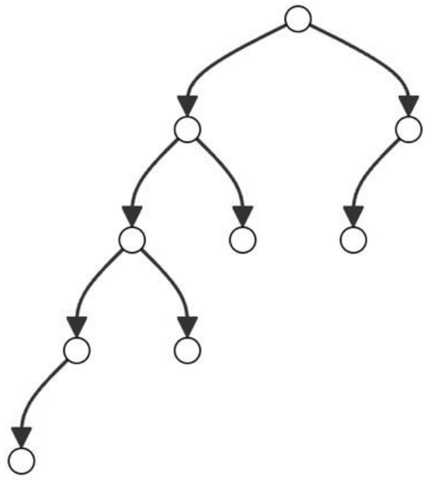
Part Three: Short Q&A (6 marks each; 24 marks in total)

1. Given a hash table, assume that only numeric values (including both integers and floats) are allowed. Explain why hash function h(x) = int(x) is NOT a good choice.

2. Suppose x is a node in a singly linked list, what does the following code do?
```
t.next = x.next
x.next = t
```
   Can it be replaced with the following code?
```
x.next = t
t.next = x.next
```

3. Given an initially empty AVL tree, insert the following node values in sequence: 10, 20, 30, 40. After each insertion, the AVL tree must remain balanced. Draw the AVL tree structure after each insertion, and identify the types of rotations that occur during the insertion process, if any.

4. Given a BST below in which L means left child, while R means right child, can it be a valid red-black tree? Explain your answer.



Part Four: Design and Analysis (36 marks in total)

(Note that pseudo-code and even plain English are allowed.)

1. Given a singly linked list below, please design an algorithm to compute the sum of positive values. For example, 3 → -2 → 1 → 4 return 8. Note that an empty linked list, or a linked list without positive values returns 0.

   class LinkedList:
```
    def __init__(self):
        self._head = None

    def positive_sums(self):
        pass # implement this
```
   (a) Please design an iterative *positive_sums*(). (7 marks)
   (b) Please design a recursive *positive_sums*(). (3 marks)

2. (a) Please assign values 0,2,4,6,8,10,12,14, 16 into the BST in Question 4, Part Three, and draw the resulting BST. (2 marks)
   (b) Given a binary tree with a root node, please design an algorithm to return the ceiling node of a given key *x*, which is the smallest node smaller than *x*. For example, the ceil of 1 is node with 2; the ceil of 4 is the node with 4; the ceil of 18 is None. (8 marks)

```
class BST:
    def __init__(self):
        self._root = None

    def ceil(self, x):
        pass # implement this
```

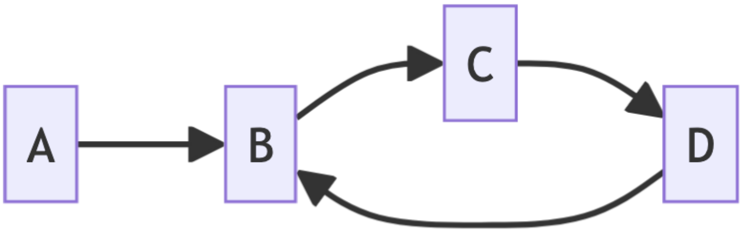
3. Given a social network in which each vertex is a user, and if two users are friends, then there is an edge between them. Each user is characterized by an id (user's indenfier). Please check whether two users can be reached with at most k hoops. (10 marks)
   Hint: It returns either True or False.

```
class SociaNetwork:
    def __init__(self, n: int): # n is the number of users
            self.n = n
            self.adjacency_list = {vertex: [] for vertex in range(n)}

    def add_friend(self, user1: int, user2: int):
            self.adjacency_list[user1].append(user2)
            self.adjacency_list[user2].append(user1)

    def at_most_khoops(self, user1: int, user2: int, k: int):
        pass # implement this
```

4. You can choose either A or B. (6 marks)
   A. Prove or disprove: f(n) = O(g(n)) implies $2^{f(n)} = O(2^{g(n)})$.
   B. A cyclic linked list is a linked list that contains at least one node whose next pointer points back to a previous node in the list, creating a loop or cycle within the list, as illustrated below. Please design an algorithm to check whether it is a cyclic linked list in O(N) time.
   Hint: It returns either True or False. An empty linked is NOT cyclic.




```
class LinkedList:
      def __init__(self, head=None):
          self._head = head
      def has_cycle(self):
          pass # implement this
```